

BASAVARAJESWARI GROUP OF INSTITUTIONS

BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT

NACCAccredited Institution*

(Recognized by Govt. of Karnataka, approved by AICTE, New Delhi & Affiliated to
Visvesvaraya Technological University, Belagavi)

"JnanaGangotri" Campus, No.873/2, Ballari-Hospet Road, Allipur,
Ballari-583 104 (Karnataka) (India)

Ph: 08392 – 237100 / 237190, Fax: 08392 – 237197



DEPARTMENT OF CSE(DATA SCIENCE)

A Mini-Project Report On

***“Interactive Football Match Performance Analysis and
Visualization”***

A report submitted in partial fulfillment of the requirements for the

DATA VISUALIZATION WITH POWER BI

Submitted By

K Ambika

USN: 3BR23CD034

**Under the Guidance of
Mr.Sundeep Sindhanoor
Asst. Professor**

***Dept of CSE (DATA SCIENCE),
BITM, Ballari***



Visvesvaraya Technological University

BASAVARAJESWARI GROUP OF INSTITUTIONS

BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT

NACCAccredited Institution*

*(Recognized by Govt. of Karnataka, approved by AICTE, New Delhi & Affiliated to
Visvesvaraya Technological University, Belagavi)*

*"JnanaGangotri" Campus, No.873/2, Ballari-Hospet Road, Allipur,
Ballari-583 104 (Karnataka) (India)*

Ph: 08392 – 237100 / 237190, Fax: 08392 – 237197



DEPARTMENT OF CSE (DATA SCIENCE)

CERTIFICATE

This is to certify that the OPEN-ENDED PROJECT of DATA VISUALIZATION WITH POWER BI “ ***Interactive Football Match Performance Analysis and Visualization***” has been successfully presented by ***K AMBIKA*** bearing USN ***3BR23CD034*** student of semester B.E for the partial fulfillment of the requirements for the award of ***Bachelor Degree in CSE(DS)*** of the BALLARI INSTITUTE OF TECHNOLOGY& MANAGEMENT, BALLARI during the academic year 2025-2026.

Signature of guide

Mr. Sundeep Sindhanoor

Signature of HOD

Dr. Aradhana D

ACKNOWLEDGEMENT

The satisfactions that accompany the successful completion of our mini project on ***“Interactive Football Match Performance Analysis and Visualization”*** would be incomplete without the mention of people who made it possible, whose noble gesture, affection, guidance, encouragement and support crowned my efforts with success. It is our privilege to express our gratitude and respect to all those who inspired us in the completion of our mini-project. We are extremely grateful to our Guides ***Mr.Sundeep Sindhanoor*** for their noble gesture, support co-ordination and valuable suggestions given in completing the mini-project. We also thank ***Dr. Aradhana D***, H.O.D. Department of CSE(DS), for his co-ordination and valuable suggestions given in completing the mini-project. We also thank the Principal, Management and non-teaching staff for their co-ordination and valuable suggestions given to us in completing the project.

NAME	USN
K Ambika	3BR23CD034

Table of Contents

ChNo	Chapter Name	Page
1		1-2
	Introduction 1.1 About the project 1.2 Language used	
2	Scope of the project 2.1 Problem statement 2.2 Objectives System requirements	3
3	3.1 Functional Requirements 3.2 Non-Functional Requirements 3.3 Hardware Requirements 3.4 Software Requirements	4-5
4	Methodology 4.1 Data Collection 4.2 Data Preparation(Power Query) 4.3 Data Visualization 4.4 Filters and Interactivity 4.5 Dashboard Design 4.6 outcome	6-8

5	Implementation 5.1 Dataset Description 5.2 Data Loading 5.3 Data Transformation(Power Query) 5.4 Data Modelling 5.5 DAX Measures(Key Calculations) 5.6 Dashboard Visualization 5.7 Filters / Slicers 5.8 Dashboard Layout 5.9 Integration 5.10 Final output	9-13
6	Result	14-16
	Conclusion	17
	References	18

ABSTRACT

The “Football Match Performance Analysis Dashboard” project is designed to provide an interactive and user-friendly visualization of football match data using Power BI. The dataset is generated using Python, simulating real-world match scenarios with details such as goals scored, goals conceded, match results, and player performance. The project transforms raw data into meaningful insights through dynamic visualizations, enabling users to analyze team performance effectively. The dashboard includes key metrics such as total matches, wins, losses, draws, goal difference, and average goals scored, which provide a quick overview of performance. Various visual elements such as bar charts, pie charts, and tables are used to represent match results, goal comparisons, and player performance levels. Interactive filters allow users to explore the data based on players and match outcomes, making the analysis more flexible and insightful. This project demonstrates how data visualization techniques can be applied in sports analytics to track performance trends, evaluate players, and support data-driven decision-making. Overall, the dashboard provides a simple, clear, and effective platform for analyzing football match performance.

CHAPTER 1

INTRODUCTION

1.1 About the Project

This project focuses on analyzing football match performance using an interactive Power BI dashboard. The dataset is generated using Python, which simulates match data such as goals scored, goals conceded, match results, and player performance.

The main objective of this project is to transform raw match data into meaningful insights using data visualization techniques. The dashboard provides a clear understanding of team performance, including win/loss trends, goal analysis, and player contributions.

By using dynamic visuals and filters, users can easily explore match statistics and identify patterns in performance. This project demonstrates how data analytics can be applied in sports to improve decision-making and performance evaluation.

1.2 Language Used

The core tools used in this project are:

Python: Used to generate the football dataset with random match statistics.

Power BI: Used for data visualization and dashboard creation.

DAX: Used to create calculated measures such as total matches, win percentage, and average goals.

CSV: Used as the data source format for importing data into Power BI.

CHAPTER 2

SCOPE OF THE PROJECT

2.1 Problem Statement

Analyzing football performance manually is difficult due to the large amount of match data. It is challenging to track team performance, compare results, and evaluate player contributions effectively.

This project solves the problem by creating an interactive dashboard that visualizes match results, goals analysis, and player performance in a simple and understandable way.

2.2 Objectives

The main objectives of the project are as follows:

- Analyze football match performance using data visualization
- Track wins, losses, and draws
- Compare goals scored and conceded
- Evaluate player performance levels
- Create an interactive dashboard with filters

CHAPTER 3

SYSTEM REQUIREMENTS

3.1 Functional Requirements

- Import dataset into Power BI
- Process match data (goals, results, players)
- Calculate metrics (wins, losses, average goals)
- Create charts and KPI cards
- Apply filters (player, result)
- Update visuals dynamically

3.2 Non-Functional Requirements

- User-friendly interface
- Fast performance
- Accurate results
- Clear and clean dashboard design
- Easy to update data

3.3 Hardware Requirements

- ***Operating System:*** Windows 10, Windows 11, or Windows Server 2016/2019/2022.
- ***Processor (CPU):*** 1.4 GHz 64-bit (x64) processor or better.
- ***Memory (RAM):*** 4 GB RAM.
- ***Storage:*** 2 GB of available hard disk space (for the Power BI Desktop application and cache).
- ***Display:*** 1440 x 900 or 1600 x 900 (16:9) resolution.

3.4 Software Requirements

- ***Development:*** Microsoft Power BI Desktop (requires Windows 10/11).
- ***Data Source:*** CSV files (can be managed via Microsoft Excel or standard text editors).
- ***Deployment & Hosting:*** Power BI Service (cloud platform for publishing and sharing the dashboard).

CHAPTER 4

METHODOLOGY

The methodology of this project follows a systematic approach to transform raw football match data into meaningful insights using Power BI. The process involves multiple stages including data collection, data preparation, data visualization, and dashboard design.

4.1 Data Collection

The dataset for this project is generated using Python. Random values are used to simulate football match data, including goals scored, goals conceded, match results, and player performance.

The dataset contains multiple match records with different players and performance levels. This approach helps in creating a realistic dataset for analysis and visualization in Power BI.

4.2 Data Preparation (Power Query)

Power Query is used to clean and prepare the dataset after importing it into Power BI.

The following steps are performed:

- Checked for missing or null values
- Renamed columns for better readability
- Ensured correct data types (numbers for goals, text for players and results)
- Verified data consistency

These steps ensure that the dataset is accurate and ready for analysis.

4.3 Data Visualization

The cleaned dataset is used to create interactive visualizations in Power BI.

The following visuals are used:

KPI Cards:

Display key metrics such as total matches, wins, losses, draws, goal difference, and average goals scored.

Bar Charts:

Used to compare goals scored and goals conceded.

Pie Chart:

Shows the distribution of match results (Win, Loss, Draw).

Column Chart:

Displays the count of matches by best performer category.

Table:

Shows detailed player performance including goals scored and performance level.

4.4 Filters and Interactivity

To make the dashboard interactive, slicers (filters) are added.

The following filters are used:

- Player
- Result (Win/Loss/Draw)

These filters allow users to dynamically explore the data and analyze performance based on selected conditions.

4.5 Dashboard Design

The dashboard is designed in a structured layout for better readability and user experience.

Top Section:

Displays KPI cards such as total matches, wins, losses, draws, and goal difference.

Middle Section:

Contains charts for goals analysis and match results.

Bottom Section:

Displays player performance table and detailed insights.

This structured layout helps users quickly understand the overall performance and analyze specific details.

4.6 Outcome

The outcome of this project is an interactive Power BI dashboard that provides clear insights into football match performance.

It helps in analyzing team results, identifying goal trends, and evaluating player performance effectively. The use of filters allows users to explore the data dynamically and gain better insights.

CHAPTER 5

IMPLEMENTATION

The implementation phase of this project focuses on transforming a vast global dataset of power generation and emissions into actionable insights through Power BI. This systematic process encompasses loading raw CSV files, executing ETL transformations in Power Query, calculating complex KPIs using DAX, and designing an interactive, user-friendly visualization layout.

5.1 Dataset Description

The dataset used in this project consists of football match data with the following fields:

Match_ID – Unique identifier for each match

Goals_Scored – Number of goals scored

Goals_Conceded – Number of goals conceded

Result – Match outcome (Win, Loss, Draw)

Player – Name of the player

Best_Performer – Performance category of the player

5.2 Data Loading

The dataset is loaded into Power BI using the “Get Data” option.

The CSV file generated using Python is selected, previewed, and then loaded into Power BI for further processing.

5.3 Data Transformation (Power Query)

Data transformation is performed using Power Query Editor.

The following steps are carried out:

- Removed any duplicate or inconsistent records
- Ensured proper formatting of columns
- Converted data types (numeric and text)
- Verified data accuracy

These transformations ensure that the dataset is clean and ready for analysis.

5.4 Data Modeling

- The project uses a single-table data model where all the data is stored in one table.
- This simplifies the analysis and allows Power BI to efficiently perform calculations and filtering across all fields.

5.5 DAX Measures (Key Calculations)

The following DAX measures are created:

- **Total Matches** = COUNT(Match_ID)
- **Wins** = CALCULATE(COUNT(Match_ID), Result = "Win")
- **Losses** = CALCULATE(COUNT(Match_ID), Result = "Loss")
- **Draws** = CALCULATE(COUNT(Match_ID), Result = "Draw")
- **Avg Goals Scored** = AVERAGE(Goals_Scored)
- **Goal Difference** = SUM(Goals_Scored) - SUM(Goals_Conceded)

5.6 Dashboard Visualizations

The dashboard includes various visualizations to represent the data clearly:

- ***KPI Cards:***
Show key metrics such as matches, wins, losses, and goal difference.
- ***Bar Charts:***
Used to compare goals scored and conceded.
- ***Pie Chart:***
Represents the proportion of match results.
- ***Table:***
Displays player-wise performance details.

These visuals help in understanding the overall performance effectively.

5.7 Filters / Slicers

Slicers are added to enhance interactivity in the dashboard.

Filters include:

- Player
- Result

These slicers allow users to customize the view and analyze specific data.

5.8 Dashboard Layout

The dashboard layout is organized as follows:

Top → KPI cards showing key performance indicators

Middle → Charts for results and goal comparison

Bottom → Player performance table

This layout ensures a clear and logical flow of information for users.

5.9 Integration

All visuals in the dashboard are connected to the same dataset.

Filters and slicers are integrated with all visuals, ensuring that selecting any filter dynamically updates the entire dashboard.

5.10 Final Output

The final output is an interactive Power BI dashboard that provides a complete analysis of football match performance.

The dashboard allows users to:

- Track match results
- Analyze goal trends
- Evaluate player performance
- Perform interactive filtering

This project successfully demonstrates how data visualization can be used to analyze sports performance effectively.

CHAPTER 6

RESULT

The developed “Football Match Performance Analysis Dashboard” successfully provides an interactive and visually clear analysis of football match data. The dashboard transforms raw match information into meaningful insights using various visualizations such as KPI cards, bar charts, pie charts, and tables.

The KPI cards display important metrics such as total matches, wins, losses, draws, goal difference, and average goals scored. These metrics provide a quick overview of the overall team performance. The bar charts are used to compare goals scored and goals conceded, helping to identify whether the team has strong offensive or defensive performance.

The pie chart represents the distribution of match results (Win, Loss, Draw), making it easy to understand the overall performance ratio. The table visualization displays detailed player performance, including goals scored and performance levels such as Star Player, Good, and Needs Improvement.

The dashboard also includes interactive filters such as player and match result, allowing users to dynamically explore the data. When filters are applied, all visuals update automatically, enabling customized analysis and better understanding of specific scenarios.

Through this dashboard, users can easily identify performance trends, evaluate player contributions, and compare match outcomes. The integration of multiple visual elements ensures that the data is presented in a simple, clear, and effective manner.

Overall, the project successfully achieves its objective of converting raw football match data into an interactive dashboard that supports data-driven analysis and decision-making in sports performance evaluation.

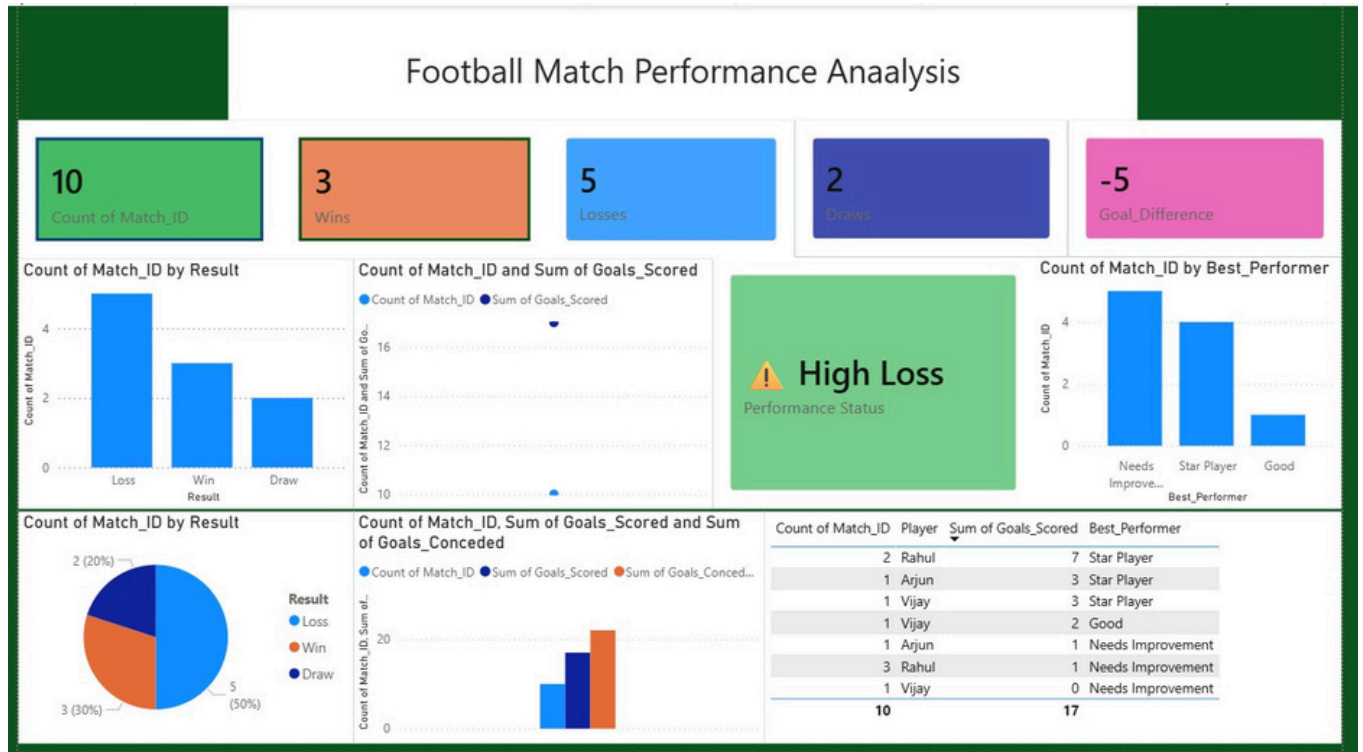


Fig 1 : Football match performance analysis

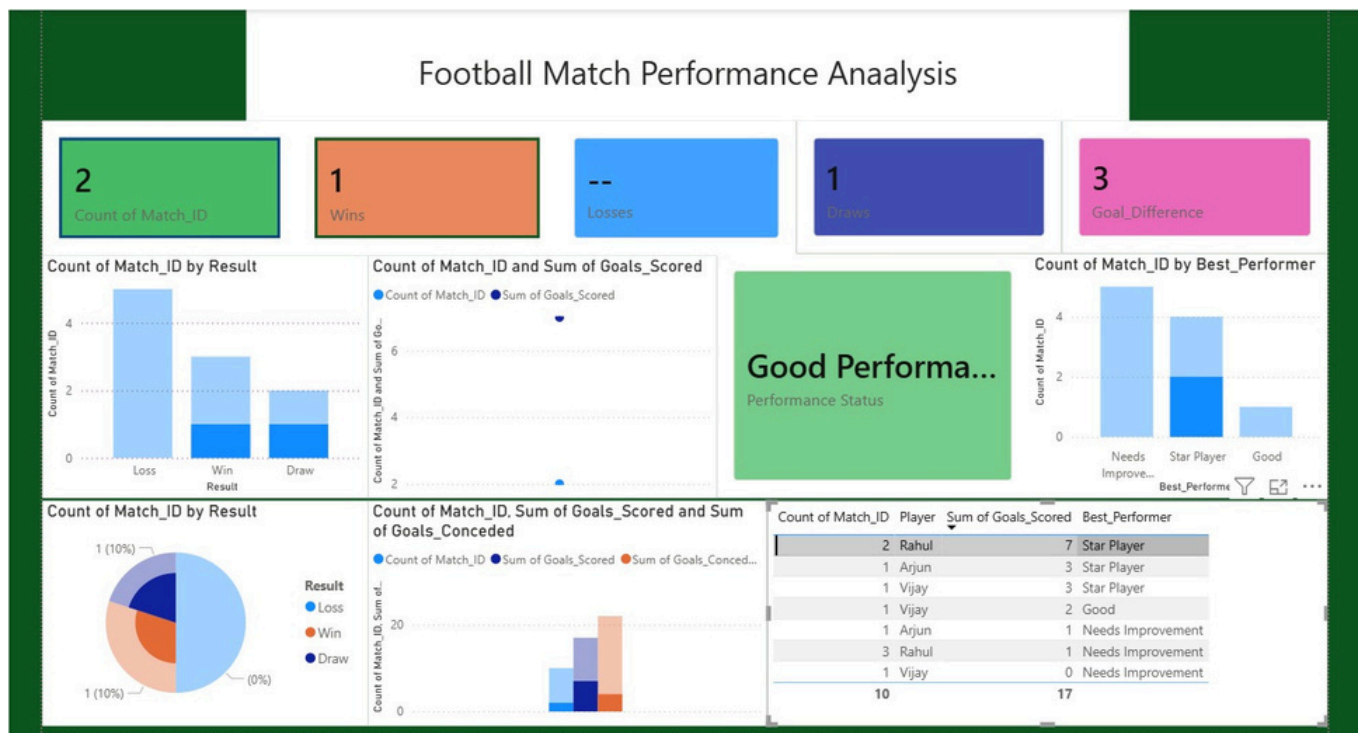


Fig 2 : Good performer of the match

CONCLUSION

The “Football Match Performance Analysis Dashboard” project successfully demonstrates the use of data visualization techniques to analyze football match data in an effective and meaningful way. By generating the dataset using Python and visualizing it using Power BI, the project converts raw data into useful insights.

The dashboard provides a comprehensive view of team performance through key metrics such as total matches, wins, losses, draws, goal difference, and average goals scored. Various visualizations like bar charts, pie charts, and tables help in understanding match results, goal trends, and player performance clearly.

The inclusion of interactive filters enhances user experience by allowing dynamic data exploration based on player and match results. This makes the dashboard more flexible and user-friendly for analysis.

Overall, the project highlights the importance of data-driven decision-making in sports analytics and demonstrates how tools like Power BI can be used to present data in a simple, clear, and interactive manner for better performance evaluation.

REFERENCES

- [1] Microsoft, “Power BI Documentation,” Microsoft Learn.
Available: <https://learn.microsoft.com/power-bi>

- [2] Microsoft, “Data Analysis Expressions (DAX) Reference,” Microsoft Learn.
Available: <https://learn.microsoft.com/dax>

- [3] Python Software Foundation, “Python Documentation.”
Available: <https://docs.python.org/3/>

- [4] Wes McKinney, “Python for Data Analysis,” O’Reilly Media, 2nd Edition.

- [5] Marco Russo and Alberto Ferrari, “The Definitive Guide to DAX,” Microsoft Press.

- [6] Microsoft, “Power Query Documentation,” Microsoft Learn.
Available: <https://learn.microsoft.com/power-query>

- [7] Online tutorials and resources on Power BI dashboard development.

- [8] Educational videos and guides on data visualization techniques.